

Chukwuemeka Ezumezu.

Sr. MLOps Engineer

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PROFILE

Chukwuemeka is a dynamic Machine Learning Engineer with solid experience in developing and deploying advanced machine learning models, data analysis, and software development. He is proficient in cloud computing, MLOps, and algorithm development, with expertise in Python, SQL, and various machine learning frameworks. He has demonstrated the ability to enhance business strategies through data-driven insights and innovative solutions. He is seeking a challenging role to leverage his skills in a mission-driven, fast-paced environment.

SKILLS

- Python
- Azure
- Machine Learning
- SQL
- LLMs

More

FastAPI, Modelling, CI/CD, GitHub, Docker, App service, Synthetic Data Generation, TensorFlow Extended, AI, SDLC, GCP, AWS, Flask, Streamlit, GitHub, Bitbucket, HTML, CSS, JavaScript, NumPy, Pandas, ETL, NLP, Experimental Design, Recommendation Systems, Linear Algebra, Plotly, Clustering, PCA, Data Processing

INTERESTS

As a constant learner, he stays updated with the latest advancements in technology and enjoys participating in tech communities and hackathons.

EXPERIENCE

In All Media | Sr. MLOps Engineer (Remote)
(Client: UL Solutions)

Austin, Texas

September 2024 - Present

Project: Developed the OPUS Medical Devices Known Problems (MDKP) service, a text classifier predicting whether medical device problems were caused by human misuse, using reports from the MAUDE database.

Contributed to Artificial Intelligent Document Extraction Version 2 (AIDE V2), an AI-powered solution for extracting structured information from documents.

Reviewed and resolved Bandit security vulnerabilities across 71 AILab repositories.

- Activities:
- Designed and implemented an end-to-end functional test for automating JSON file processing through Azure Blob Storage and Inference pipelines. Implemented data visualization solutions for real-time reporting and analysis.
 - Configured workflows to move files between Landing Zone containers and inference data stores using object replication, ensuring file integrity and tagging during processing.
 - Automated file output delivery to Landing Zones and integrated REST API notifications for successful file transfers.
 - Developed parsing services for document extraction, splitting layouts into structured chunks (by page, size, etc.), and validated functionality through local and CI testing.
 - Built CI pipelines with code quality checks, ensuring 80%+ code coverage, and implemented markdown conversion for document layouts.
 - Reviewed 71 AILab repositories to resolve Bandit security vulnerabilities, integrated Bandit checks into CI pipelines, and ensured security compliance.
 - Automated inference workflows, reducing manual intervention and improving processing efficiency.
 - Improved document extraction accuracy through modular parsing services and robust testing.
 - Strengthened repository security by identifying and resolving vulnerabilities and embedding security checks into CI pipelines.

Technologies: Python, Azure ML Studio, Azure DevOps (ADO), Azure MLOps SDK, Azure Python SDK, Azure CLI, Azure Blob Storage, GitHub, CI/CD pipeline

Nurovant AI | AI/ML Engineer (Remote)

Dallas, Texas

March 2024 - August 2024

Project: Utilized ASR techniques with the NVIDIA NeMo model to conduct Speaker-Diarization tasks, transcribing audio files into speaker segments and timestamps. Developed and integrated the Teacher Speak Rate feature in Flask, analyzing diarized transcripts to optimize educators' speaking time and enhance student engagement during lectures.

The software is used to transcribe audio files into speaker segments and timestamps, allowing for detailed analysis of speaking patterns. The Teacher Speak Rate feature is designed to monitor and optimize educators' speaking time, enhancing student engagement during lectures. The project focused on improving educational outcomes through advanced speech recognition and analytics.

- Activities:
- Developed and integrated the Teacher Speak Rate feature in Flask, improving student engagement.
 - Implemented the Questions Tracker with LLM to monitor classroom questions, fostering educational dialogue.
 - Created the GPTZero-powered content classifier feature to differentiate AI-generated content.
 - Developed an ML-driven document proofreader feature to enhance document quality.
 - Implemented CI/CD pipelines for continuous deployment of ML models in production environments.
 - Automated the drift monitoring and retraining of ML models to detect and correct data drift.
 - Enhanced student engagement with advanced audio analysis features.
 - Improved academic integrity with content classification tools.
 - Continuously optimized models and workflows with CI/CD pipelines using GitHub Actions.
 - Increased system robustness with automated monitoring and retraining processes, reducing model degradation by 25%.

Technologies: Python, Flask, FastAPI, NVIDIA NeMo, SQL, AWS (EC2, S3 Bucket), Azure App Service, GitHub Actions, Docker

Chekkit Technologies Corp | Lead ML Engineer

Ikoyi, Lagos

September 2022 - March 2024

Project: Developed and deployed ChekIntel, a real-time business intelligence web application, to Azure App Service, resulting in a 10% profit increase. Designed and implemented ChekInspect, a computer vision solution for product inspection and quality assurance.

ChekIntel is used to provide real-time business intelligence by aggregating and analyzing data, helping businesses make informed decisions that drive profitability. ChekInspect is utilized for product inspection and quality assurance through computer vision technology, ensuring that products meet quality standards before reaching customers. Both applications focused on enhancing operational efficiency and product quality through advanced data analysis and computer vision techniques.

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EXPERIENCE

Activities:

- Designed and implemented ChekInspect for product inspection using computer vision.
- Built predictive models for user survey data analysis, enhancing decision-making.
- Developed and deployed ChekIntel on Azure App Service, driving an increase in profits.
- Integrated ML workflows with Azure DevOps pipelines for efficient model and version management.
- Implemented cloud resource optimization strategies to improve performance and reduce costs.
- Increased profits by 10% through successful deployment of ChekIntel.
- Improved product quality assurance with ChekInspect.
- Enhanced operational efficiency with Azure DevOps pipeline integration, reducing deployment time by 40%.
- Reduced cloud operational costs by 15% through resource optimization and model management.

Technologies: Python, Azure, Tableau, SQL, Machine Learning, Flask, Azure DevOps, Kubernetes

Freelance | Machine Learning Engineer Lagos Island, Lagos
June 2017 - February 2024

Project 1: Developed a Retrieval Augmented Generation (RAG) powered PDF and website links chatbot app using LangChain, ChatGPT API, Huggingface language models, and Python Flask. Also developed a mobile breaking news app with over 1000 users on the Play Store, generating revenues. The RAG-powered chatbot app is designed to interact with users, providing accurate and contextually relevant information from PDFs, SQL database, and website links by leveraging advanced language models and Flask. The mobile breaking news app delivers real-time news updates to its users, offering an engaging and user-friendly interface that has successfully attracted a significant user base and generated steady revenue. Both projects focused on enhancing user experience through innovative applications of AI and mobile technology.

Activities:

- Developed a RAG-powered chatbot app for PDFs and website links using LangChain and Huggingface.
- Created a mobile breaking news app with over 1000 users, generating revenue.
- Configured CI/CD pipelines for continuous deployment and integration of language models in the chatbot app.
- Developed an automated performance monitoring system for model fine-tuning in the news app.
- Significantly enhanced user interaction with the RAG-powered chatbot app.
- Achieved user growth and revenue with the mobile breaking news app.
- Reduced deployment and model update time in the chatbot app by 30% with CI/CD pipelines.
- Improved chatbot performance by 20% with automated hyperparameter tuning.

Technologies: Python, Flask, LangChain, ChatGPT API, Huggingface, Flutter, GitHub Actions, Elasticsearch

Project 2: Contributed to a project focused on preventing malaria infections through topography and satellite image analysis. The project aimed to prevent malaria infections by analyzing topographic data and satellite images to identify and monitor stagnant water bodies, which are breeding grounds for mosquitoes. By leveraging advanced image analysis techniques, the project sought to provide actionable insights for targeted interventions, ultimately reducing the prevalence of malaria in affected areas.

Activities:

- Developed an algorithm for identifying stagnant water bodies, contributing to malaria prevention.
- Implemented an automated monitoring system for continuous satellite image analysis and detection of new risk areas.
- Automated the generation of risk area reports to facilitate intervention decision-making.
- Developed an efficient algorithm for identifying stagnant water bodies, preventing mosquito breeding.
- Significantly contributed to reducing malaria prevalence through automated monitoring.
- Improved intervention decision-making speed with automated risk area reporting.

Technologies: Python, OpenCV, PyTorch, scikit-learn, NumPy, Pandas

Public Complaints Commission | Sr. Data Analyst Lagos Island, Lagos
September 2013 - September 2022

Project: Conducted research, analyzed data using statistical techniques, identified trends, prepared reports, and made recommendations for the commission. The research project involved leveraging statistical techniques to analyze collected data, identify significant trends, and derive actionable insights. The comprehensive analysis and reporting aimed to support the commission in making informed decisions, improving service delivery, and enhancing operational efficiency. The recommendations provided were based on thorough data analysis, ensuring they were well-founded and impactful.

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EXPERIENCE

- Activities:
- Developed a new filing system for proactive cases, improving case file management.
 - Implemented data visualization solutions for real-time reporting and analysis.
 - Automated data collection and analysis processes using Python scripts, reducing significantly the processing time.
 - Developed a data pipeline for integrating multiple sources and real-time database updates.
 - Increased complaint closure rate by 30% through data-driven decisions.
 - Improved service delivery efficiency by 28% with a new filing system.
 - Reduced data processing time by 40% through automation of scripts.
 - Enhanced decision-making speed with real-time data integration pipelines.

Technologies: Python, Data Analysis, Statistical Techniques, SQL, ETL

EDUCATION

- University of Sunderland, UK
MSc Data Science (In Progress)
2025 - 2026
- National Open University of Nigeria (NOUN)
Computer Science
2014 - 2019

- Certifications:
- AI Product Manager Nanodegree
 - Intel Edge AI for IoT Developers Nanodegree
 - Deep Learning Nanodegree
 - Machine Learning Engineer Nanodegree